METRIC

DOD-S-51517
22 April 1986
SUPERSEDING
O-S-377b
April 20, 1966

MILITARY SPECIFICATION

SODIUM PYROPHOSPHATE, ANHYDROUS, TECHNICAL (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of anhydrous sodium pyrophosphate which is also known as tetrasodium phyrophosphate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-35 - Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated
PPP-D-723 - Drums, Fiber
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons)

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research and Development Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6810

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STANDARDS

FEDERAL

Fed. Std. No. 313 - Material Safety Data Sheets Prepraration and the Submission Of
FED. TEST METHOD STD. NO. 536/GEN - Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are indicated as DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM STANDARDS

D 460 - Sampling and Chemical Analysis of Soaps and Soap Products
D 1193 - Reagent Water (DOD Adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Appearance. Anhydrous sodium pyrophosphate shall be a white, uniform material in either granular or powdered form when tested as specified in 4.2.4.1.

3.2 Chemical and physical characteristics. Anhydrous sodium pyrophosphate shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Requirement	Minimum	Maximum	Test paragraph
Assay as $Na_4P_2O_7$, percent by weight	98.0		4.2.4.2
Matter insoluble in water, percent weight		0.2	4.2.4.3
Loss on ignition, percent by weight		0.5	4.2.4.4
Iron content, parts per million		70	4.2.4.5
Particle size: passing a No. 16 sieve, percent by weight	99		4.2.4.6

3.3 Material Safety Data Sheets. Material Safety Data Sheets for anhydrous sodium pyrophosphate shall be prepared and submitted by the contractor as specified in Fed. Std. No. 313 (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the anhydrous sodium pyrophosphate offered for acceptance at one time, which have been produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.4).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For anhydrous sodium pyrophosphate tests. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 300 grams (g) shall be removed from each sample container and placed in a suitable, dry, sealed container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for anhydrous sodium pyrophosphate tests

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit container, ready for shipment. Sample unit containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible

4.2.3.2 For anhydrous sodium pyrophosphate tests. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

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4.2.4.1 Appearance. Visually examine the specimen for color, uniformity, and form.

4.2.4.2 Assay. Determine percent by weight assay as $Na_4P_2O_7$ in accordance with the procedure for Tetrasodium Pyrophosphate in ASTM D 460.

4.2.4.3 Matter insoluble in water. Determine the percent by weight matter insoluble in water in the specimen in accordance with the procedure for Matter Insoluble in Water in ASTM D 460.

4.2.4.4 Loss on ignition. Weigh 2 g of the specimen to the nearest milligram into a porcelain crucible which has previously been ignited to constant weight. Heat in a mufle furnace at 700°C for 30 minutes, cool to room temperature in a desiccator and weigh to the nearest milligram. Calculate the percent by weight loss on ignition as follows:

$$\text{Loss on ignition} = \frac{100A}{W}$$

where: A = Loss in weight in grams and
W = Weight of specimen in grams.

4.2.4.5 Iron.

(a) Standard iron solution. Dissolve about 2.0 g of ferric chloride ($FeCl_3 \cdot 6H_2O$) in a liter of water containing 10 milliliters (mL) of concentrated hydrochloric acid. Determine the ferric iron content of this solution by a standard method. From a buret, run the calculated volume of this solution into a 1-liter volumetric flask to yield a concentration of 0.002 milligram of iron per milliliter. Dilute to the volume mark with water. One mL of the solution is equivalent to 20 parts per million of iron in a 0.1 g specimen.

(b) Procedure. Weigh 1.00 g of the specimen into a 250-mL beaker and add 15 mL of nitric acid having a specific gravity of 1.42 and 5 mL of sulfuric acid having a specific gravity of 1.84. Heat in a hood until dense fumes of sulfur troxide are evolved and long enough to destroy any organic matter that might be present. Cool to room temperature; add 30 mL of water, 10 mL of concentrated hydrochloric acid, and 0.3 mL of nitric acid having a specific gravity of 1.42; boil for 10 minutes; cool to room temperature; and transfer to a 100-mL volumetric flask. In case the iron content is low (10 to 20 parts per million), use the whole specimen and dilute to 50 mL with acetone for the determination. Dilute to the volume mark with water, shake well, and transfer a suitable aliquot to a 50-mL Nessler tube. Add 25 mL of acetone and 5 mL of concentrated hydrochloric acid, mix well using a clean stopper, and allow to stand for 10 minutes. Add 10 mL of 20-percent ammonium thiocyanate solution and after mixing, compare with freshly prepared standards, made by treating aliquots of the standard iron solution prepared in (a) in the manner described above and finally diluting to the same volume as the specimen. Any color produced in the specimen solution shall be no darker than that produced in the standard solution.

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4.2.4.6 Particle size. Determine the percent by weight specimen passing a No. 16 sieve in accordance with method 536/2101 of FED. TEST METHOD STD. NO. 536/GEN.

5. PACKAGING

5.1 Unit packing. Anhydrous sodium pyrophosphate shall be unit packed level A, B, or C as specified (see 6.2).

5.1.1 Level A. A quantity of 45.0 (+0.5 or -0) kilograms (kg) of anhydrous sodium pyrophosphate shall be unit packed in a pail conforming to type II of PPP-P-704 or in a drum conforming to type II, grade A of PPP-D-723. Pails and drums shall have interior coatings which will not have a deleterious effect on the containers or their contents.

5.1.2 Level B. A quantity of 45.0 (+0.5 or -0) kg of anhydrous sodium pyrophosphate shall be unit packed in a bag conforming to type III, class 1, grade III, style B, bag No. P6 of PPP-B-35.

5.1.3 Level C. A quantity of 45.0 (+0.5 or -0) kg of anhydrous sodium pyrophosphate shall be unit packed to ensure carrier acceptance and safe delivery to destination in a container complying with the rules and regulations applicable to the mode of transportation.

5.2 Packing. The unit packs shall also serve as the shipping containers for all levels of packing.

5.3 Unitization. When specified (see 6.2), unit packs shall be palletized in accordance with MIL-STD-147.

5.4 Marking. Containers shall be marked in accordance with MIL-STD-129. In addition, each container shall be marked to show the lot number and date of manufacture of the anhydrous sodium pyrophosphate.

6. NOTES

6.1 Intended use. Anhydrous sodium pyrophosphate is intended for use in various washing, cleaning, and scouring processes with soap and other detergents where a mildly alkaline material with appreciable water softening properties is desired. There may be local restrictive use of this material for certain applications; guidance on local use should be obtained from the site-specific environmental office.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of unit packing required (see 5.1)
(c) If palletization is required (see 5.3)

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6.3 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with Fed. Std. No. 313. The pertinent mailing addresses for submission of data are listed in appendix B of Fed. Std. No. 313.

6.4 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.6 Supersession data. This specification supersedes and includes the requirements of O-S-377b, dated April 20, 1966.

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD
DLA - CS

User activity:

Navy - SH

Preparing activity:

Army - EA

Project No. 6810-B539

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-S-51517	SODIUM PYROPHOSPHATE, ANHYDROUS, TECHNICAL (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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